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Bereken $\int \frac{4}{\sqrt{x-5}} dx$

Substitutie: $t = \sqrt{x-5}$, $t^2 = x-5$, $dx = 2t dt$

$$\begin{aligned} \int \frac{4}{\sqrt{x-5}} dx &= \int \frac{4}{t} \cdot 2t dt \\ &= 8 \int dt \\ &= 8t + c \\ &= 8\sqrt{x-5} + c \end{aligned}$$

References